

Bayesian Food Recommendation System

Abstract

Food recommendation systems often struggle to adapt to evolving user preferences and typically provide recommendations without indicating the confidence of their predictions. This project proposes a **Hybrid Bayesian Personalized Meal Recommendation System** that combines Bayesian inference with collaborative filtering to generate adaptive and personalized recipe recommendations. The system predicts the probability that a user will enjoy a recipe based on factors including available ingredients, dietary restrictions, cuisine preferences, cooking time, nutritional information, previous recipe ratings, and similarities to other users. Unlike traditional recommendation systems, the Bayesian component continuously updates user preferences through posterior probability distributions as new interactions occur, allowing recommendations to become increasingly personalized while quantifying prediction uncertainty. By combining collaborative filtering with Bayesian updating, the proposed system aims to improve recommendation accuracy while providing interpretable confidence estimates for each recommendation.

Method

The proposed system models recipe recommendation using a **hybrid Bayesian recommendation framework** that combines **collaborative filtering** with **Bayesian inference**. First, a collaborative filtering model identifies recipes that a user is likely to enjoy by learning patterns from historical user–recipe interactions. We plan to use the **Singular Value Decomposition (SVD)** algorithm implemented in the **Surprise** library to generate initial recommendation scores based on similarities between users and recipes.

The Bayesian component then refines these recommendations by updating prior probability distributions into posterior distributions as users provide new ratings and feedback. Recipe features such as ingredients, cuisine type, cooking time, nutritional information, dietary tags, and historical ratings are incorporated into this updating process. The final recommendation score combines collaborative filtering predictions with Bayesian posterior probabilities and provides a confidence score that quantifies the uncertainty of each recommendation. This hybrid methodology is well suited for the project because collaborative filtering captures patterns across similar users, while Bayesian inference enables continuous personalization as user preferences evolve and provides interpretable confidence estimates.

Pipeline:

User Ratings



Collaborative Filtering (SVD): Learns from all users' rating patterns



Initial Recipe Predictions



Bayesian Updating: Updates predictions using the user's evolving preferences and uncertainty



Final Personalized Recommendation

Model

The project will be implemented in **Python** using the **Surprise** library for collaborative filtering and **PyMC** for Bayesian modeling and inference. Specifically, the **SVD algorithm** from the Surprise library will generate initial personalized recommendations based on user–recipe interactions, while PyMC will perform Bayesian updating to refine recommendations as new user feedback becomes available. This approach does not rely on a pre-trained language model but instead combines established recommendation algorithms with Bayesian inference to create a personalized and continuously learning recommendation system.

Example:

Suppose you're a new user.

You've rated:

- ★★★★★ Butter Chicken
- ★★★★★ Garlic Naan
- ★ Curry Salad

A **pure Bayesian** model can infer:

- Likes Indian food

- Doesn't like salads

But it **doesn't know** what thousands of other users with similar tastes also enjoyed.

A **hybrid model** can do this:

Users who liked Butter Chicken and Garlic Naan also loved:

- Chicken Tikka Masala
- Paneer Butter Masala
- Garlic Parmesan Naan

Then the Bayesian layer adjusts those recommendations based on your specific preferences (e.g., maybe you prefer vegetarian dishes or shorter cooking times).

SVD answers: *"Based on millions of ratings from everyone, what recipes is this user likely to enjoy?"*

Bayesian updating answers: *"Given this specific user's recent behavior and preferences, how should we adjust those predictions, and how confident are we?"*

Data

The project will use the **Food.com Recipes and User Interactions** dataset available on Kaggle. This real-world dataset contains over **180,000 recipes** and approximately **700,000 user ratings and reviews**, making it well suited for personalized recommendation tasks. It includes structured recipe metadata such as ingredients, nutritional information, cooking instructions, cuisine and dietary tags, as well as historical user–recipe interactions. These features provide sufficient information for collaborative filtering while also supporting Bayesian updating of user preferences as additional ratings are collected.

Dataset:

https://www.kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions?select=RAW_recipes.csv

The dataset includes:

- Over 180,000 recipes
- Approximately 700,000 user ratings and reviews
- Recipe ingredients
- Nutritional information
- Cooking instructions and preparation steps
- Recipe tags (e.g., vegetarian, dessert, Italian, gluten-free)
- Historical user–recipe interaction data for personalized recommendation